

CLAIM - Alpha as a 21|30 Closure Gate

Alpha as a closure margin from one named gate, not a fitted decimal.

This note claims that the fine-structure constant can be read as a closure margin produced by a specific 21|30 gate. The claim is conditional on a C30 carrier and a 21-step readout window; the non-trivial part is that this pair forces the rest of the ledger. Reading a 21-window on a C30 carrier forces a three-chain split, the rho=10 decagon cycle, the beat closure 210, and the first full active closure 1260.

structure

$$\alpha_{RT} = \frac{\pi}{30 \cdot 1260} (3 \cdot 30 - \Delta)$$

correction grammar

$$\Delta = AB + \frac{AB}{\rho} - \frac{1}{\rho G} - \frac{D}{\rho GCS}$$

expanded expression and bindings

$$\Delta = 2 + \frac{2}{10} - \frac{1}{10 \cdot 39} - \frac{20/21}{10 \cdot 39 \cdot 30 \cdot 7}$$

$$AB = 2, \quad \rho = 10, \quad G = 3(14 - 1) = 39, \quad D = \frac{20}{21}, \quad C = 30, \quad S = 7$$

numerical comparison (CODATA not used as input)

$$\alpha_{RT} = 0.0072973525630485$$

$$\Delta(\alpha^{-1}) \approx 2.41 \times 10^{-8}$$

$$\alpha_{RT}^{-1} = 137.035999201092$$

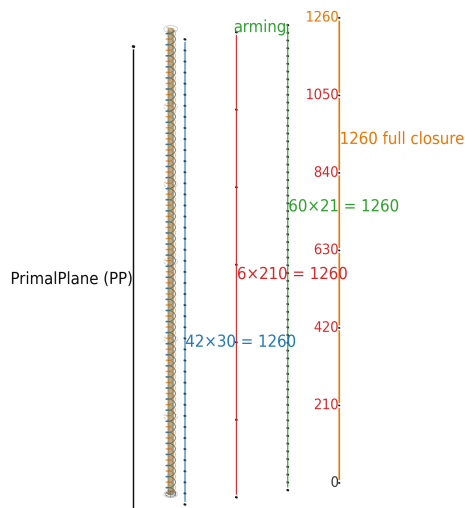
$$\alpha_{CODATA\ 2022}^{-1} = 137.035999177(21)$$

Gate value computed from the 21|30 ledger. CODATA is shown only as an external comparison, not as input.

Full closure

$$1260 = 21 \times 3 \times 20 = 6 \times 210 = 42 \times 30$$

First N: active part of each 21-window is (21 - 1)=20.
Three chains: N/21 = 3 x 20 = 60, so N = 1260.



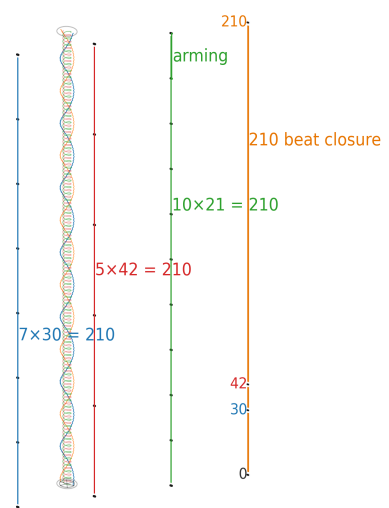
Bars and gradation ticks retained: 42 x 30, 6 x 210, 60 x 21.

Beat closure

$$210 = 7 \times 30 = 5 \times 42 = 10 \times 21$$

210 is the local beat: lcm(21,30)=210.

10 windows: 210/21=10; 7 C30 legs close with 5 forty-two waves.



Local cell where the 21|30 gate becomes visible.

Break condition

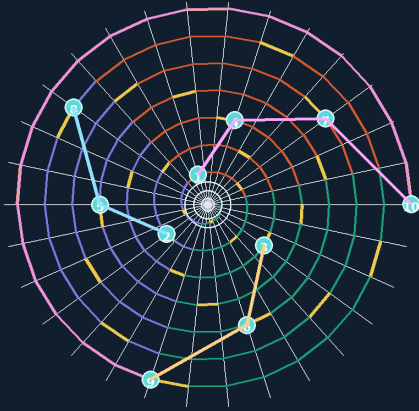
Identify a link invalid, non-unique under stated rules, or replaceable by a stricter construction.

THE 21|30 GATE - Break the Ledger

The point is not that the numbers are interesting; the point is that the links co-lock.

One beat: the 21|30 gate

Intentionally non-physical display. Geometry is unfolded for readability.
Visual aid only; the claim stands or falls on the ledger links.



full spiral visible: $210 = 10 \times 21 = \text{lcm}(21,30)$

W21 on C30

read a 21-window on a 30-slot carrier

$\text{gcd}(21,30) = 3$

forced split into three transport chains

$30 / 3 = 10$

rho / decagon readout, 36 degrees per chain step

$\text{lcm}(21,30) = 210$

one local beat closure

$N = 21 \times 3 \times 20 = 1260$

full closure: 3 chains x (21-1) active positions

$G = 3(14 - 1) = 39$

42 = 3 x 14, then one seam excluded per chain

Companion visualizer (not part of the claim): package file RT_ALPHA_READOUT_GEOMETRY.html; GitHub path: https://github.com/Bo-Fremling/rhythm-theory-release/blob/Release/RT_ALPHA_READOUT_GEOMETRY.html. It is only an interactive reader aid; the claim stands or falls on the ledger links below.

Gate link	Consequence
$3 = \text{gcd}(21,30)$	forced split into three transport chains
$10 = 30 / 3$	rho = 10 / decagon readout
$210 = \text{lcm}(21,30)$	one local beat closure
$1260 = 21 \times 3 \times 20 = 6 \times 210 = 30 \times 42$	first full active closure: 3 x (21 - 1) active positions
$42 = 1260 / 30 = 3 \times 14$	C30 carrier turns under the three-chain split
$G = 3 \times (14 - 1) = 39$	one excluded seam per transport chain
$D = (21 - 1) / 21 = 20/21$	active readout duty: one P-ARM / boundary slot reserved

Correction grammar, active duty, and type rule

$$\Delta = AB + AB/\rho - 1/(\rho G) - D/(\rho G C S)$$

Allowed factors forced by the 21|30 gate: $AB=2$, $\rho=10$, $G=39$, $D=20/21$, $C=30$, $S=7$.

Expansion: $\Delta = 2 + 2/10 - 1/(10 \times 39) - (20/21)/(10 \times 39 \times 30 \times 7)$.

Active duty D

$N = Z3 * |L_{\text{cap}}| = 3 * 7 = 21$
one P-ARM / boundary slot is reserved
active readout = 20, hence $D = 20/21$

Type rule

each term is measured in its own domain
chain quotient: counting measure -> numerator 1
full 21|30 readout: readout mask -> numerator D

Therefore the numerator schema is AB, AB, 1, D. A stricter competing grammar must preserve the same domain/measure typing.

Final claim

alpha_RT is proposed as a dimensionless closure margin generated by the named 21|30 gate. Scope: this note concerns only the alpha claim; the HTML viewer is a reader aid. To refute the claim, break a named gate link, the active-duty derivation, the type rule, or provide an equal-or-stricter competing grammar from another primitive gate. Free recombinations do not count.